FROB C5 quotient endomorphism controls

Validation audit, 22 October 2026

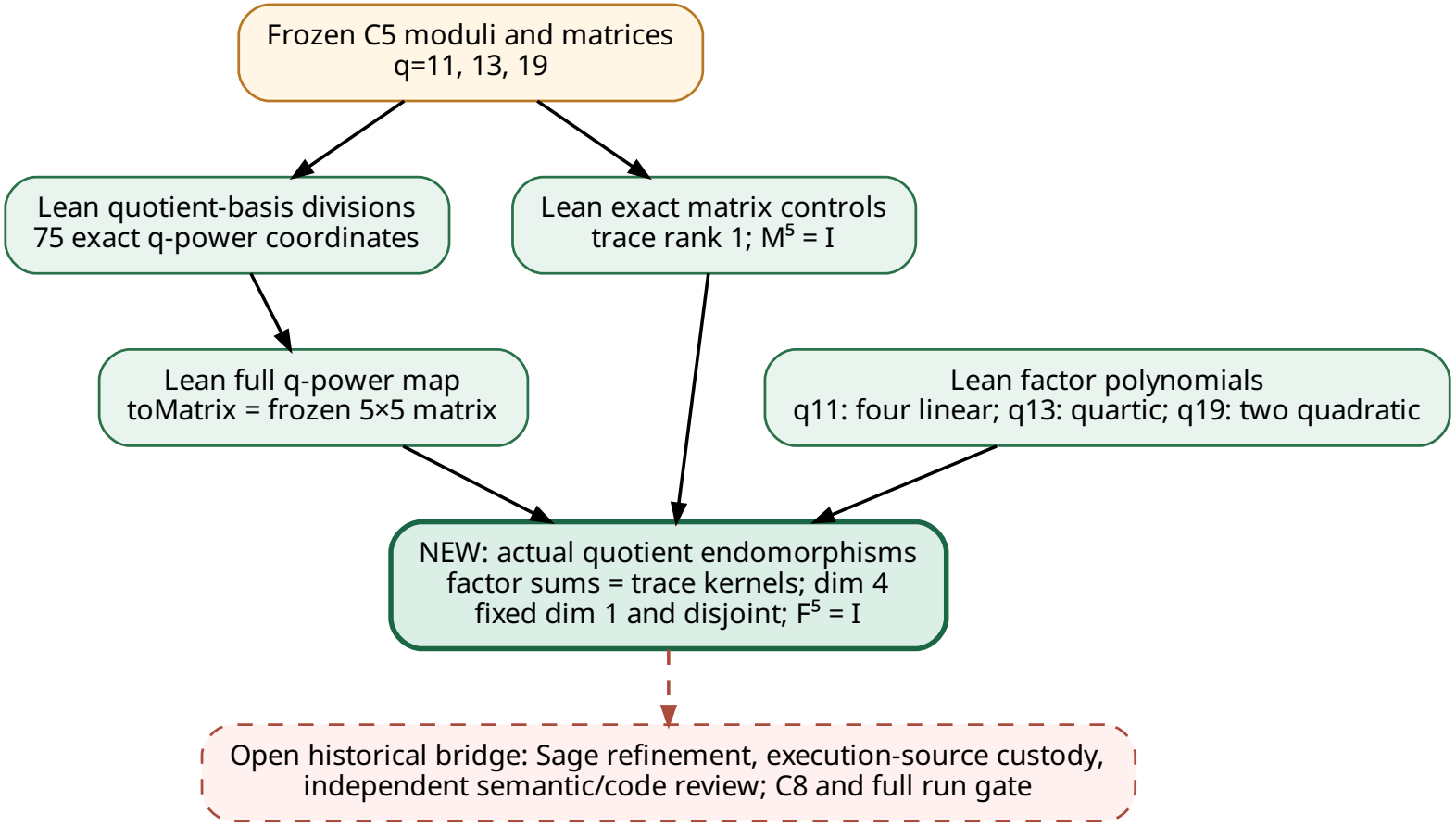
Scope. Lean proves C5 factor-kernel, trace, fifth-power and fixed-line controls for the actual q-power endomorphisms of the three frozen polynomial quotient algebras. These are ring-level quotient theorems; irreducibility of the quotient moduli is not assumed or established in Lean.

Base field	Frozen quotient modulus	Checked endomorphism controls
$\mathbf{F}_{11}$	$z^5 + 2z^4 + 1$	four factor kernels, trace dim 4, fixed dim 1
$\mathbf{F}_{13}$	$z^5 + 8z^4 + 1$	quartic factor kernel, trace dim 4, fixed dim 1
$\mathbf{F}_{19}$	$z^5 + 3z^4 + 1$	two factor kernels, trace dim 4, fixed dim 1

Proof path. Prior Lean certificates prove all 75 frozen q-power matrix entries and the full linear-map equality in each power basis. An algebra equivalence from endomorphisms to reindexed matrices transports the exact matrix trace rank 1 and $M^5 = I$ facts. The abstract polynomial factor identities then apply to the actual q-power maps. Rank-nullity gives trace-kernel dimension 4 and fixed-kernel dimension 1; a fixed vector in the trace kernel is zero because 5 is nonzero in each base field. The aggregate target theorem checks all three cells and their factor-kernel directness/equality, fifth-power, dimension and disjointness statements.

Machine verification. Source commit `20982d9a84d8` uses pinned Lean 4.35.0-rc1. The exact-source verifier passed an 8,976-job full build and 3,099-job target build, found no forbidden constructs, and audited 686 theorem declarations with standard axioms only. The requested theorem's axiom audit passed. The JSON receipt records `machine_verified`; independent semantic review remains pending.

Boundary. The frozen Sage driver computes the same categories of matrix and kernel flags, and a separate source-bound Python replay checks the archived C5 cells. The driver's `T_equals_hM` flag compares two constructions of the same matrix polynomial and is only a consistency check. Historical Sage-to-Lean refinement and original execution-source custody have not been established. Modulus irreducibility in Lean, C8, the complete experiment gate and performance claims remain open. No scientific trial or timing run was repeated.


Frozen C5 moduli and matrices
q=11, 13, 19
Lean quotient-basis divisions
75 exact q-power coordinates
Lean exact matrix controls
trace rank 1; M^5 = I
Lean full q-power map
toMatrix = frozen 5×5 matrix
Lean factor polynomials
q11: four linear; q13: quartic; q19: two quadratic
NEW: actual quotient endomorphisms
factor sums = trace kernels; dim 4
fixed dim 1 and disjoint; F^5 = I
Open historical bridge: Sage refinement, execution-source custody,
independent semantic/code review; C8 and full run gate